

# Productivity Model Training & Evaluation Report

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## 1. Dataset Summary

| Metric             | Value                               |
|--------------------|-------------------------------------|
| Total Records      | 100,000 records                     |
| Features Count     | 20 features                         |
| Train / Test Split | 80% (80,000) / 20% (20,000)         |
| Target Variable    | productivity_score (0.00 to 100.00) |
| Target Mean Score  | 55.24                               |
| Target Range       | 0.00 to 100.00                      |

## 2. Model Comparison Table

| Model Name                  | R <sup>2</sup> Score | RMSE   | MAE    | Explained Var |
|-----------------------------|----------------------|--------|--------|---------------|
| Linear Regression           | 0.9424               | 4.8420 | 3.8534 | 0.9424        |
| Random Forest Regressor     | 0.9482               | 4.5924 | 3.6509 | 0.9482        |
| Gradient Boosting Regressor | 0.9489               | 4.5604 | 3.6190 | 0.9489        |

## 3. Best Model Highlights

| Attribute                    | Value                              |
|------------------------------|------------------------------------|
| Selected Best Model          | Gradient Boosting Regressor        |
| Highest R <sup>2</sup> Score | 0.9489                             |
| Lowest RMSE                  | 4.5604                             |
| MAE                          | 3.6190                             |
| Saved Model File             | models/productivity/best_model.pkl |
| Saved Scaler File            | models/productivity/scaler.pkl     |